Product Kernel Perspective Spaces: Predicting Benchmark Scores from Sparse, Unpaired Model Evaluations

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Abstract

Evaluating language models is expensive, and in practice benchmark coverage is *doubly sparse and noisy*: a model is run on only a subset of tasks, within a task on only a subset of queries, and the resulting per-cell score is a noisy average over those few queries. Methods that complete the (model $\times$ task) score matrix—low-rank matrix completion, item-response theory—use only the *aggregate* score of each cell. We argue that the cheapest way to evaluate better is to use *item-level* information: not just a cell's score but its individual responses. We introduce the **Product Kernel Perspective Space (PKPS)**, which embeds each model from its responses through a product kernel $k_Q \cdot k_R$ that compares responses to *similar* queries (not only identical ones), recovering the Data Kernel Perspective Space as the bandwidth $\sigma \to 0$ and degrading gracefully when models answer different queries. PKPS supplies an item-level *embedding* signal; ensembling it with the available item-level *score* signal—a cell's own sample mean where it was measured, matrix completion where it was not—recovers the full score matrix from observations where each cell has any number of queries (efficient benchmarking and benchmark completion become the two ends of one problem). We characterize *when* the embedding helps: given the same preprocessing as matrix completion, an effective-rank analysis shows the embedding *matches* completion at every rank on low-rank matrices and *beats* it when the matrix is high-rank (as on MATH), so effective rank sets the size of its advantage; it also survives unpaired queries where DKPS and IRT collapse, and rescues the small-cohort regime where low-rank completion fails for lack of rows. Across three observation levers—task coverage, queries per cell (noise), and cohort size—the single embedding ensemble beats the best score-only predictor on a heterogeneous 18-task, 93-model suite spanning math, translation, medical QA and legal classification, by a margin that grows as evaluation gets sparser, noisier, or smaller-cohort. Item-level information pays wherever real evaluation is incomplete, which is everywhere that matters.

1 Introduction

Comprehensive evaluation of language models is a growing bottleneck: leaderboards track hundreds of models across dozens of benchmarks, yet no model is run on everything. The resulting (model $\times$ task) score matrix is *doubly sparse*. *Across tasks*, a model is evaluated on a subset of benchmarks (missing tasks). *Within a task*, evaluation may use only a subset of queries to save cost (missing queries). Predicting the missing entries—a model's score on a task it has not been (fully) run on—would let practitioners reuse cached evaluations instead of paying to re-run them [Polo et al., 2024, Vivek et al., 2024, Zeng and Papailiopoulos, 2026].

The Data Kernel Perspective Space (DKPS) [Anonymous, 2026] is a black-box approach: it embeds each model from its query responses, places models in a low-dimensional Euclidean space via classical multidimensional scaling, and regresses benchmark scores in that space. DKPS is query-efficient and competitive with item-response theory. But it has a structural limitation: the distance between two models is the average squared difference of their responses *to the same queries*, so DKPS requires a *paired* design. When models answer different queries—the common case in real, organically-assembled leaderboards—DKPS has nothing to compare and the information in those unpaired responses is thrown away.

We introduce the **Product Kernel Perspective Space (PKPS)**. The key change is to compare responses to *similar* queries, not only identical ones, by weighting each response-pair comparison with a query kernel. This (i) lets every unpaired response contribute, and (ii) contains DKPS as the special case where the query kernel is the identity. The broader thesis is that PKPS supplies an *item-level embedding* signal that score-completion baselines ignore, and that combining it with the available *item-level score* signal predicts benchmark performance more efficiently than either alone. Our contributions:

- **Method.** A product-kernel distance $k_Q \cdot k_R$ that generalizes DKPS, reduces to it as the query-kernel bandwidth $\sigma \to 0$, and whose bandwidth is chosen per run by a cheap median-scaled cross-validation (Section 3). We treat it as the embedding term of an embedding-$\oplus$-score ensemble.
- **Unified estimation problem.** We recover the full (model $\times$ task) matrix from observations where each cell has $m \geq 0$ queries, with one ensemble of the own sample (m>0), matrix completion (m=0), and the PKPS embedding (everywhere), structured by three levers—task coverage, queries per cell (noise), and cohort size (Section 4.1).
- **Where the embedding helps.** With the same preprocessing as matrix completion, an effective-rank analysis shows the embedding *matches* completion at every rank on low-rank matrices and *beats* it where the matrix is high-rank (MATH), so rank sets the size of its advantage; it survives unpaired queries where DKPS and IRT collapse; and it rescues the small-cohort regime where rank-2 completion fails for lack of rows (Section 4.2).
- **One ensemble, every lever.** On a heterogeneous 18-task, 93-model suite spanning math, translation, medical QA and legal classification, the embedding already beats matrix completion on its own across coverage and cohort, and the single ensemble beats the best score-only predictor across noise, coverage, and cohort—by a margin that grows as evaluation gets sparser, noisier, or smaller-cohort (Section 4.2).

2 Related work

Low-dimensional representations of models. A recurring empirical finding is that the behavior of a language model is governed by a handful of latent factors, so a model can be summarized by a short vector. Burnell et al. [2023] recover three interpretable factors (reasoning, comprehension, core language modeling) that explain 82% of the across-model variance on HELM [Liang et al., 2023] by factor analysis; Ruan et al. [2024] show that downstream performance across $\sim$100 models is well described by a low-dimensional *capability space*, with families differing mainly in how efficiently compute is converted into those capabilities; and item-response theory [Rasch, 1960, Polo et al., 2024, Kipnis et al., 2024] places each model on a latent ability axis fit from item-level correctness. These all consume *aggregate* per-task scores. The Data Kernel Perspective Space (DKPS) [Anonymous, 2026] instead builds a *black-box, response-level* representation: it embeds each model from its cached query responses via classical multidimensional scaling [Torgerson, 1952] and regresses scores in that space. PKPS is of this response-level kind but removes DKPS's requirement that every model answer the *same* items, so the representation is defined even on organically-assembled, unpaired evaluations.

Benchmark prediction. A second line predicts unobserved (model, task) scores so that not every evaluation must be run; *efficient benchmarking* and *matrix completion* are its two ends. Efficient benchmarking lowers the per-task cost by scoring a curated item subset and extrapolating—IRT-based item selection [Polo et al., 2024, Kipnis et al., 2024], anchor points [Vivek et al., 2024], reliability-driven subset design [Perlitz et al., 2024], and response-space prediction from cached

items [Anonymous, 2026]—while reuse of past evaluations on a shared sample set is its cross-model analogue [Prabhu et al., 2024]. Matrix completion instead predicts cells that were *never* run by exploiting low rank across the (model × task) matrix: classical low-rank imputation [Candès and Recht, 2009, Mazumder et al., 2010, Koren et al., 2009] and, most directly, BenchPress [Zeng and Papailiopoulos, 2026], which finds the frontier-model score matrix is approximately rank-two and completes it in logit space with calibrated confidence. Both ends use only the *aggregate* score of each cell and discard the item-level responses. PKPS unifies them—a cell observed with $m \geq 0$ queries spans efficient benchmarking ($m>0$) and completion ($m=0$)—and supplies an item-level *embedding* signal complementary to the score-only view, which we ensemble with matrix completion rather than replace.

3 The Product Kernel Perspective Space

Models and observations. A *model* is a random mapping $f : \mathcal{Q} \to \mathcal{X}$ from a query space $\mathcal{Q}$ to a response space $\mathcal{X}$; querying f at q returns a response drawn from the distribution $f(q)$. We observe n models $f_1, \ldots, f_n$, each run on its own *query set* $Q_i \subseteq \mathcal{Q}$, giving a tuple $\mathbf{Q} = (Q_1, \ldots, Q_n)$ that may differ across models or be disjoint; the paired design $Q_1 = \cdots = Q_n$ is the special case. Two fixed, distinct embedding functions map raw objects to vectors: a *response* embedding $\text{embed}_R : \mathcal{X} \to \mathbb{R}^p$ and a *query* embedding $\text{embed}_Q : \mathcal{Q} \to \mathbb{R}^{p'}$ (e.g. different text-embedding models). For $q_j \in Q_i$ we average the embeddings of the r sampled responses, $x_{ij} = \frac{1}{r}\sum_{k=1}^{r} \text{embed}_R\big(f_i(q_j)^{(k)}\big) \in \mathbb{R}^p$, and write $u_j = \text{embed}_Q(q_j)$ for the embedded query. The goal is to predict each model's benchmark score $y_i \in \mathcal{Y}$.

Perspective space. Each method summarizes the n models in an $n \times n$ affinity matrix A (defined below). The squared distance between two models is $D^2(a,b) = A_{aa} + A_{bb} - 2A_{ab}$ (clamped at 0), and their d-dimensional *representations* are the classical-multidimensional-scaling minimizer

$$(\hat{\psi}_1, \ldots, \hat{\psi}_n) = \operatorname*{arg\,min}_{z_1,\ldots,z_n \in \mathbb{R}^d} \sum_{i,k=1}^{n} \big(\|z_i - z_k\| - D(i,k)\big)^2, \tag{1}$$

so model i is the point $\hat{\psi}_i \in \mathbb{R}^d$, its *perspective*. Paired DKPS and PKPS differ only in the affinity A.

Paired DKPS. With all n models answering the same m queries ($Q_1 = \cdots = Q_n$), DKPS uses the identity-matched affinity $A_{ab} = \frac{1}{m}\sum_j k_R(x_{aj}, x_{bj})$: response j of model a is paired only with response j of model b, so the design must be paired.

Product kernel. PKPS replaces identity matching with a query kernel k_Q:

$$A_{ab} = \frac{\sum_{j \in Q_a}\sum_{l \in Q_b} k_Q(q_j, q_l)\, k_R(x_{aj}, x_{bl})}{\sum_{j \in Q_a}\sum_{l \in Q_b} k_Q(q_j, q_l)}, \tag{2}$$

where Q_a is the query set of model a. Each term is normalized by its own query-kernel mass, so models that answered different numbers of (or entirely different) queries are still comparable; responses to *similar* queries—even across tasks—contribute and unpaired evaluations are used (Figure 1). Paired DKPS is the special case $k_Q = \delta$, where only $j = l$ survives.

Kernels. The response kernel k_R compares embedded responses, the query kernel k_Q compares embedded queries. Paired DKPS uses a linear response kernel and the identity query kernel $k_Q = \delta$. Throughout our analysis we use a linear or RBF k_R and an RBF query kernel over the embedded queries, $k_Q(q_j, q_l) = \exp(-\|u_j - u_l\|^2/2\sigma^2)$, with bandwidth σ chosen per run (below).

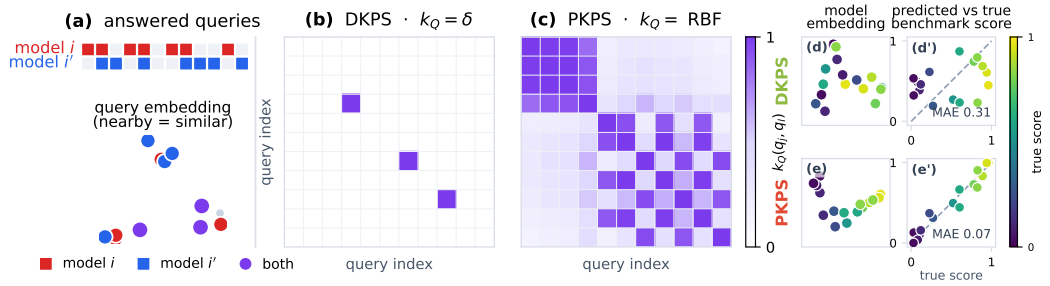

Figure 1: Why the query kernel matters under *unpaired* evaluation: PKPS can use all available information. **Left:** two models, i (red) and i' (blue), each answer their own subset of a shared query pool, shown both as answered-query blocks (top) and as points in query-embedding space (bottom; nearby points are similar queries). The two models share only a few queries. Both kernel panels index the queries by a single universal ordering (shared by rows and columns). **Center (DKPS, $k_Q = \delta$).** The identity query kernel keeps a cell only when the two queries are the *same* ($q_j = q_l$, the diagonal) *and* both models answered it ($q_j \in Q_i \cap Q_{i'}$), so only the few shared queries survive—almost every response is discarded. **Right (PKPS, k_Q = RBF).** The RBF query kernel over the same ordering is a dense, *symmetric* similarity matrix, so responses to *similar* queries—not only identical ones—bridge the two models and every response can contribute. PKPS thus turns the sparse shared-query diagonal DKPS depends on into a dense similarity kernel. **Right block (d, d′, e, e′):** a benchmark-prediction example on a small model population. Each model answers only a few queries from a large pool, so two models rarely answer the *same* query though they answer *similar* ones. Each method embeds the models by classical MDS of the pairwise distance matrix (**d, e**: the 2-D model representations, coloured by true score and rotated so the score gradient runs left-to-right), then predicts each model's full-benchmark score by leave-one-out regression onto that embedding (**d′, e′**: predicted vs. true). Restricted to the rare shared queries, **DKPS** yields a noisy geometry with no clean score axis and predictions scatter off the diagonal (MAE 0.31); **PKPS** bridges all similar queries, so score varies smoothly across its embedding and predictions land on the diagonal (MAE 0.07).

3.1 Theoretical properties of the PKPS

PKPS is a strict generalization of DKPS. **(i) DKPS limit.** Take $k_Q = \delta$, the indicator $\mathbf{1}[q_j = q_l]$. Since the double sum already ranges over $j \in Q_a$, $l \in Q_b$, a term survives only if the two queries are equal *and* that query lies in both sets, $\mathbf{1}[q_j = q_l]\mathbf{1}[q_j \in Q_a]\mathbf{1}[q_l \in Q_b] = \mathbf{1}[q_j = q_l \in Q_a \cap Q_b]$, so $A_{ab} = \frac{1}{|Q_a \cap Q_b|}\sum_{q \in Q_a \cap Q_b} k_R(x_{aq}, x_{bq})$ —exactly paired DKPS over the shared queries (the same-m-queries case recovers $\frac{1}{m}\sum_j k_R(x_{aj}, x_{bj})$). Moreover, as $\sigma \to 0$ the RBF kernel converges to δ (distinct queries have distinct embeddings), so DKPS is recovered in the small-bandwidth limit. **(ii) Unpaired comparability.** Because each affinity is normalized by its own query-kernel mass $\sum_{jl} k_Q(q_j, q_l)$, two models that answered different—or entirely disjoint—query sets remain comparable, whereas paired DKPS, which sums only over shared queries, is undefined when no query is shared. **(iii) Bandwidth interpolation.** Increasing σ smoothly trades exact query matching for cross-query bridging, so a single scalar controls how much unpaired evidence the perspective uses.

3.2 Inference in the PKPS

Inference is a two-step statistical problem, exactly as in the DKPS [Anonymous, 2026]: embed each model into the perspective space by the scaling of Eq. 1, then regress its benchmark score on the resulting coordinate. Concretely, on a set of reference models with known scores $\{(f_i, y_i)\}$ we fit a decision function $\hat{h} : \mathbb{R}^d \to \mathscr{Y}$ on the pairs $\{(\hat{\psi}_i, y_i)\}$ and predict any model's score as $\hat{y}(f) = \hat{h}(\hat{\psi}(f))$. The only free choices are the query-kernel bandwidth σ and the regressor $\hat{h}$, which

we describe next; we then add the score channel that turns the embedding into an ensemble.

Choosing the bandwidth. The optimal σ depends on how paired the data is: small when query overlap is high (recover exact matching), large when overlap is low (bridge across queries). We set it per run by cross-validation over a median-scaled grid $\sigma \in \{0.03, \ldots, 3\} \cdot \text{med}$, where med is the median pairwise query distance, selecting the σ whose perspective best predicts the *observed* scores under leave-one-model-out k-NN. The criterion never touches the held-out scores we report, so the choice is not circular with the evaluation; and because the grid spans down to near-delta, cross-validation *recovers DKPS* when query overlap is high and bridges only when it must. One response-kernel pass produces the entire σ-grid, so tuning is nearly free. We use a single regressor (distance-weighted k-NN) and a logit-space, bias-decomposed head throughout.

Estimation pipeline. For each replicate we (i) reduce response and query embeddings by PCA on the *observed* rows only (dimension by the second Zhu–Ghodsi elbow); (ii) form D from Eq. 2; (iii) embed models by classical MDS; (iv) map the embedding to scores with the *same* head as the matrix-completion baseline up to the factor model—logit transform, per-task standardization to unit variance, and a two-way (model + task) additive bias—then a k-NN regression of the residual on the embedding (replacing completion's low-rank factor), predicting every cell (held-out cells directly, observed cells leave-one-out so the estimate never sees its own target). Matching the per-task standardization is important: without it the model offset is dominated by the highest-logit-variance tasks, which silently handicaps the embedding on heterogeneous, multi-scale suites.

The embedding-$\oplus$-score ensemble. For each cell we combine the embedding with whichever score signal carries independent information. On a *missing* cell, that is matrix completion of the (noisy) score matrix [Zeng and Papailiopoulos, 2026]; we blend it with PKPS using a weight fit by held-out cross-validation against the observed scores. On an *observed* cell, the cell's own sample mean is the score signal (matrix completion merely echoes it there); we shrink it toward the PKPS estimate by a precision weight $\sigma_q^2/(\sigma_q^2 + e_{PKPS}^2)$—the sample's noise variance over its total disagreement with PKPS—so the own sample dominates as queries grow and the embedding dominates as noise grows. Both quantities are read off the observed queries—the sample noise variance estimated as $\hat{p}(1-\hat{p})/m$ and the prior error from its depth-weighted disagreement with the sample—so the weight never touches the held-out full scores and the score baselines see exactly the same information.

3.3 Query efficiency

PKPS is designed to reach a target prediction accuracy from fewer evaluated queries than paired DKPS: by bridging responses to similar queries it uses evidence that DKPS, limited to shared queries, discards. We state the expected gain as a conjecture and leave a formal analysis—in the style of the DKPS query-efficiency bound [Anonymous, 2026]—to future work; Section 4.2 is its empirical counterpart.

Conjecture 1 (Query efficiency of PKPS)**.** *Fix a benchmark score function that is Lipschitz on the perspective space and a target error ε. For a query design $\mathbf{Q} = (Q_1, \ldots, Q_n)$ with cross-model overlap ρ (the fraction of query-kernel mass shared across models), the number of queries per model that PKPS needs to reach error ε is at most that of paired DKPS, and the gap is increasing as $\rho \to 0$; the two are closest in the fully paired limit $\rho = 1$.*

4 Experiments

A real evaluation matrix is governed by four observation levers, which structure our experiments. The first is the one DKPS cannot handle: (i) *pairing*—the fraction ρ of each cell's queries that are shared across models, from fully paired (ρ=1) to fully unpaired (ρ=0). The other three set how sparse and noisy the observations are: (ii) *task coverage*—the fraction of (model, task) cells run at all; (iii) *queries per cell*—how many queries a run uses, which sets the per-cell measurement *noise*; and (iv) *cohort size*—the number of models. The all-tasks-covered, few-query limit is efficient benchmarking (denoising) and the zero-query limit is benchmark completion—two applications of the same embedding.

4.1 Synthetic data

We generate n models with latent ability vectors v_i and T tasks with latent vectors t_k, scores $v_i \cdot t_k + \varepsilon$, and queries drawn near each task center. Each model answers M_{ij} queries per task, of which $m_{ii'}$ are shared with another model (overlap $\rho = m_{ii'}/M_{ij}$); a task is observed with probability p_{task} and a query with probability p_{query}. We predict held-out (model, task) scores and compare DKPS (identity query kernel) against PKPS (RBF), with the score-only sample baseline (predict a held-out cell by its task's observed mean) as a floor (Figure 2).

PKPS—selecting its query bandwidth by the same leak-free cross-validation we use on real data—beats DKPS across *every* lever, and both beat the sample floor: bridging responses to similar queries (across tasks as well as within) supplies model-similarity signal that exact-match DKPS cannot reach, so the margin is wide even where queries are mostly paired (panels a–c). The gap is largest where pairing is scarce: as query coverage p_{query} drops (d) DKPS degrades while PKPS stays flat, and sweeping the overlap ρ directly (e), DKPS collapses toward the sample floor as $\rho \to 0$ while PKPS is essentially flat—it never needed pairing.

Panel (f) is the *denoising* face of the same problem: each cell is now *observed* with M_{ij} noisy queries (sample noise $\propto 1/\sqrt{M_{ij}}$) and we predict its true score. The raw sample mean (gray) falls like $1/\sqrt{M_{ij}}$; smoothing those noisy scores over the perspective denoises well below it while measurement noise dominates—PKPS reaches at a single query the accuracy the raw sample needs roughly five queries to match, and does so whether queries are paired or not, while DKPS denoises only when paired (unpaired DKPS, dashed, barely improves). Only once the budget is large is the raw sample precise enough to overtake the embedding's residual bias. This is the empirical face of Conjecture 1.

4.2 Real data

We evaluate on one realistic, heterogeneous benchmark, studying the two applications the unpaired embedding buys: making evaluation more *query-efficient* (denoising cheap, few-query runs) and completing *missing* cells. Both are *unpaired* regimes—each model answers its own queries, so DKPS's identity matching and IRT's shared-anchor fit degrade while PKPS bridges similar queries. All real-data numbers are means over 16 seeds, and PKPS selects its query bandwidth by cross-validation per run. The suite is 18 tasks across four datasets with very different response types—MATH (7 math subjects), WMT-14 (5 translation directions), MedQA (medical multiple-choice), and LegalBench (5 legal classification subsets)—on the 93 models evaluated on all four. MATH and WMT have rich free-text responses; MedQA and LegalBench responses are single tokens for which text embeddings are degenerate, so we encode each dataset's responses natively (Gemini embeddings vs. one-hot answers) in disjoint blocks. With a linear k_R this makes cross-dataset response similarity exactly zero, and a single shared query embedding with a within-domain bandwidth keeps k_Q near zero across domains: the product kernel factorizes by domain, never comparing a proof to a legal label.

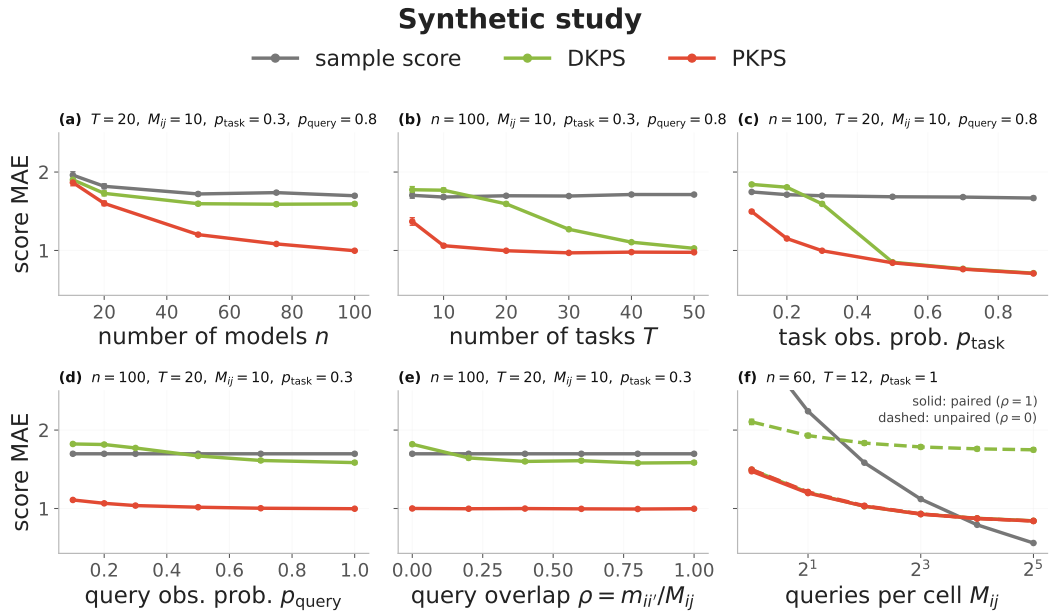

Figure 2: Synthetic study (score MAE, ± 1 SEM over 50 seeds). DKPS (identity query kernel, green) vs. PKPS (RBF, red), with the score-only sample baseline (gray); each panel sweeps one parameter with the rest fixed (shown in its title). **(a)–(e) complete held-out cells** from observed responses—cohort n, tasks T, task coverage p_{task}, query coverage p_{query}, and cross-model overlap $\rho = m_{ii'}/M_{ij}$. PKPS (CV-selected bandwidth) beats DKPS throughout—by a wide margin even when mostly paired (a–c), since it bridges across tasks as well as within—and the gap is largest when unpaired or sparse (d, e), where in (e) DKPS collapses toward the floor as $\rho \to 0$ while PKPS stays flat. **(f) denoises observed cells**: each cell is measured with M_{ij} noisy queries and we predict its true score, at paired (ρ=1, solid) and unpaired (ρ=0, dashed). The raw sample falls $\propto 1/\sqrt{M_{ij}}$; smoothing over the perspective denoises well below it at small budgets—PKPS paired or unpaired, DKPS only paired—until the sample becomes precise at the full budget.

Application 1: query efficiency. Because the embedding reads response *style*, which is stable even when the per-cell sample mean is noisy, it denoises cheap evaluations. Each model answers m queries per cell (sampled independently—so this is itself an unpaired regime, and IRT cannot fit); we predict the true full-eval score (Figure 3). At a tiny budget the embedding is far more accurate than the raw mean: at m=1, PKPS scores 0.136 and the ensemble 0.133 versus the sample's 0.292; PKPS at one query roughly matches the sample at four. As the budget grows the raw mean catches up and eventually overtakes the embedding's residual bias (0.034 vs. 0.088 at m=32); the leak-free ensemble tracks the better channel to within a few thousandths. The advantage is robust across cohort size and task coverage (panels b, c). Per cell, the denoising win is broad but tail-driven: the ensemble beats the raw sample on most cells, with the largest gains at the smallest budgets and on the rich-response datasets (Figure 4).

Application 2: matrix completion. The same embedding also predicts cells that were *never run*. We observe a fraction p_{task} of cells (each at depth p_{query}) and predict the missing ones, against a low-rank matrix-completion baseline (the sample does not apply—a missing cell has no sample). The contrast is the cohort: low-rank completion needs many rows, so when models are scarce it fails for lack of them while the embedding, needing none, holds up—at n=10, completion is 0.153 versus PKPS's 0.127 (Figure 5, dashed). At full cohort (n=93, solid) PKPS slightly edges completion (0.104 vs. 0.107) and the ensemble is best (0.100). So even on its own turf—a low-rank, dense leaderboard—completion is matched or slightly beaten; the embedding's decisive advantage is the

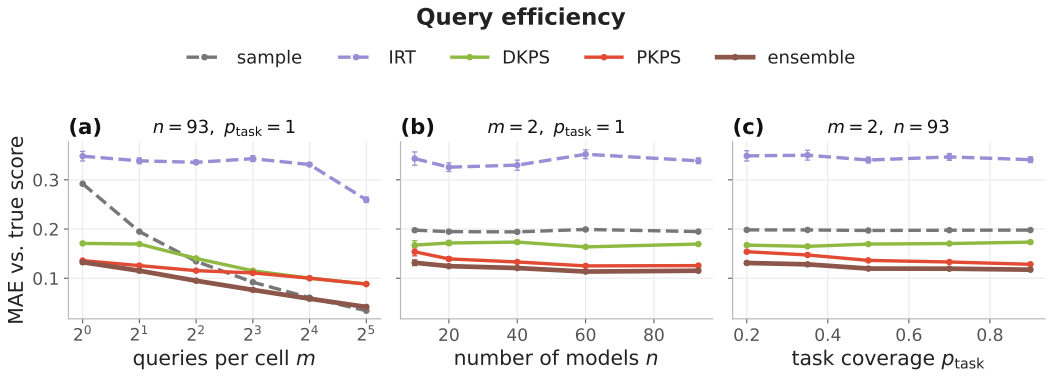

Figure 3: **Query efficiency.** Each model answers m unpaired queries per *observed* cell; predict its true full-eval score (MAE, ± 1 SEM). PKPS (red) and the ensemble (brown) denoise far below the raw sample (grey) at small budgets and across cohort size (b) and coverage (c); DKPS (green) trails because the queries are unpaired and IRT (purple) cannot fit at all.

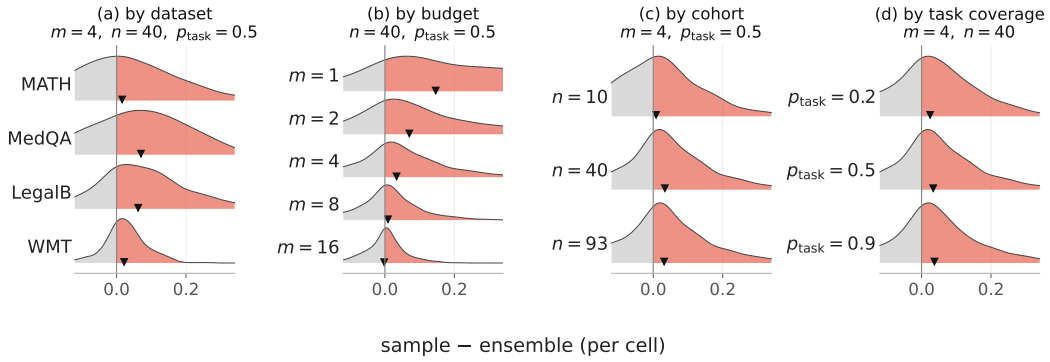

Figure 4: **Query efficiency, conditionally: where the denoising win comes from.** Per-cell improvement Δ = sample − ensemble MAE ($\Delta > 0$, red, the ensemble wins) stacked by dataset, by per-cell budget m, by cohort n, and by task coverage p_{task}, holding the other levers at their sweep median (m=4, n=40, p_{task}=0.5). The win is broad but tail-driven—largest at the smallest budgets and on the rich-response datasets—with the per-row mean (▼) positive throughout.

sparse-cohort corner completion cannot reach. The per-cell win over completion is modest and tail-driven—completion is strong on this low-rank suite—but stays positive in the mean across datasets, cohorts, coverage, and query depth (Figure 6).

Summary. Table 1 collects the two applications at a hard operating point each. Both are unpaired regimes, so the same product-kernel embedding that DKPS and IRT cannot match denoises cheap evaluations and completes missing cells—PKPS is present and competitive in both, and the ensemble, which folds in a cell's own sample, is best wherever that sample exists.

5 Discussion

Real leaderboards are assembled organically: models are run on whatever query subsets their authors chose, so the responses that benchmark estimation must compare are *unpaired*. DKPS and item-response models need a shared block of identical queries and collapse without it. PKPS removes

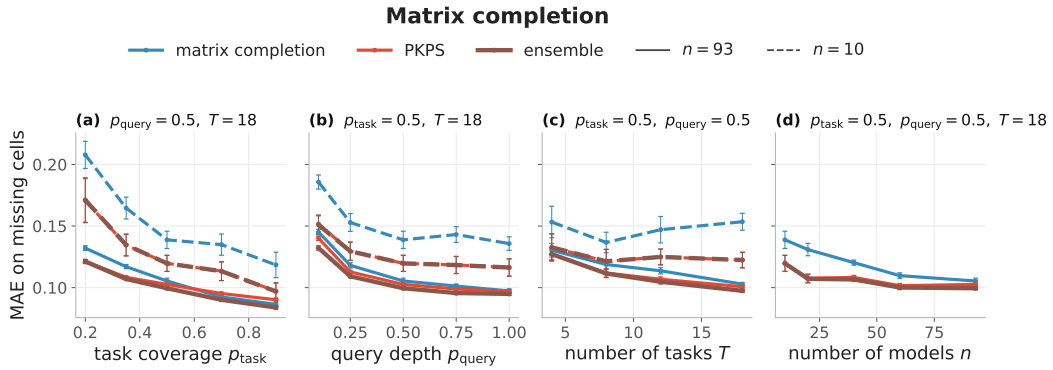

Figure 5: **Matrix completion.** Observe a fraction of (model, task) cells, predict the *missing* ones (MAE on missing cells, ± 1 SEM). Colour is method (low-rank completion blue, PKPS red, ensemble brown); line style is cohort (solid n=93, dashed n=10). The embedding rescues the small-cohort regime where low-rank completion collapses for lack of rows, and ties it at full cohort.

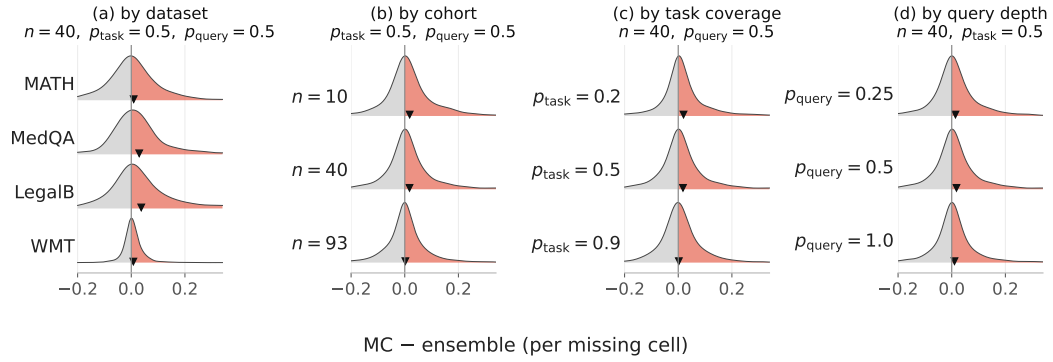

Figure 6: **Matrix completion, conditionally: where the win comes from.** Per-missing-cell improvement Δ = matrix-completion − ensemble MAE ($\Delta > 0$, red, the ensemble wins) stacked by dataset, by cohort n, by task coverage p_{task}, and by query depth p_{query}, holding the other levers at their sweep median (n=40, p_{task}=0.5, p_{query}=0.5). The win is modest and tail-driven on this low-rank suite—completion's home turf—but the per-row mean (▼) stays positive across conditions.

that requirement with a product kernel $k_Q \cdot k_R$ that compares responses to *similar* queries, recovering paired DKPS exactly in its $\sigma \to 0$ limit and scaling to a heterogeneous 18-task suite through a block-diagonal kernel. The central result is that evaluation survives unpairing: as the shared fraction $\rho \to 0$, DKPS and IRT fall off a cliff while PKPS holds. That capability buys two practical applications from the same embedding—it denoises cheap, few-query evaluations (query efficiency) and predicts cells that were never run (matrix completion), and ensembled with a cell's own sample where one exists it is never the worse choice. None of this assumes pairing or exhaustive per-cell evaluation, which is exactly what organically-assembled leaderboards cannot provide.

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Table 1: Suite MAE vs. the full-eval score at one hard operating point per regime (lower is better; best in bold). "—" marks a baseline that does not apply to that regime: the sample mean and IRT need a query a cell never ran (completion), and matrix completion needs a cell that was never run (the observed-cell regimes). PKPS is the throughline.

Method	**Query-efficient** (m=1/cell)	**Completion** (missing, n=10)
sample mean	0.292	—
IRT	0.348	—
matrix completion	—	0.153
DKPS	0.171	—
PKPS	0.136	**0.127**
ensemble	**0.133**	**0.127**

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